Key Features

Real-Time Battery Monitoring

Battery status tracking with live cloud updates for dependable flight awareness.

Temperature Monitoring

AERO-GUARD

GPS Tracking

Thermal tracking to help detect overheating and protect onboard systems.

Altitude Monitoring

Precise location monitoring for improved navigation and flight visibility.

Flight Duration Monitoring

Continuous altitude insights to support stable and controlled operation.

Motor Status

Live flight-time tracking to help manage missions and battery usage.

Project Overview

AERO-GUARD is an intelligent drone monitoring platform that continuously tracks battery level, temperature, altitude, GPS location, motor status, and flight duration using ESP32, ThingSpeak Cloud, and a web dashboard.

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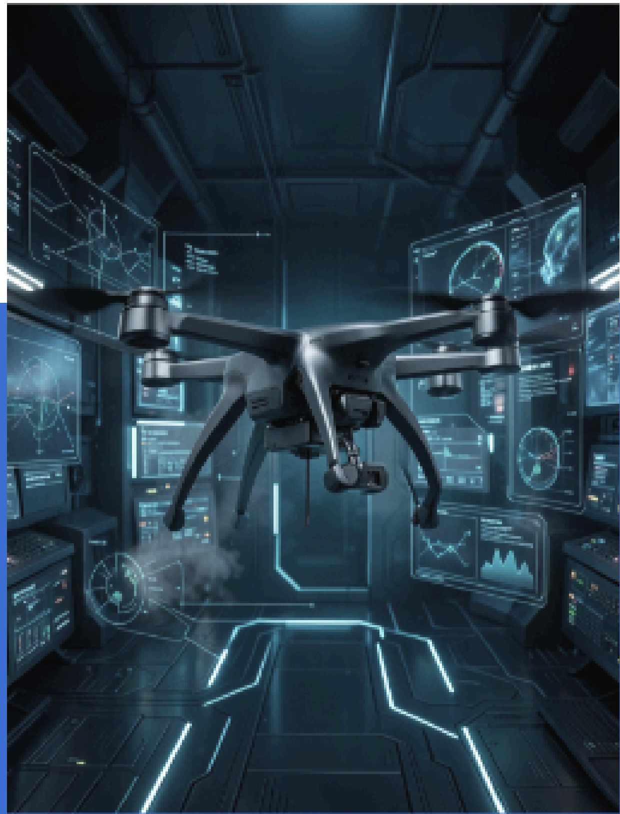
AERO-GUARD

Connecting smarter flights for safer operations.

AERO-GUARD

Smart Drone Health Monitoring System

"Real-Time Monitoring | AI Analytics | Safer Flights"



Technologies Used

ESP32, Wokwi Simulator, ThingSpeak Cloud, HTML, CSS, JavaScript, AI Analytics, and Cloud Computing power the AERO-GUARD platform.

> ***Empowering Safety, Redefining Security with AI Innovation.***

System Architecture

1

ESP32 to Sensors

Data is captured from sensors on the drone and processed by the ESP32 controller.

2

Wi-Fi to ThingSpeak Cloud

Telemetry is transmitted securely to the cloud for remote monitoring and storage.

3

Dashboard to AI Analytics

The web dashboard visualizes drone health data and AI models generate intelligent insights.

4

User Alerts

Operators receive real-time warnings, recommendations, and predictive maintenance guidance.

5

Real-Time Battery Monitoring

AI battery health prediction and smart recommendations for safer flight decisions.

6

Overheating Alerts

Instant alerts notify users when critical temperature thresholds are exceeded.

Benefits

Improved Drone Safety

Predictive Maintenance

Continuous monitoring helps reduce risk and improve operational reliability.

Real-Time Cloud Monitoring

Intelligent Flight Decisions

Live telemetry and AI insights support faster, more informed decisions.

Reduced Risk of Failure

Safer Flights

Predictive alerts and diagnostics help prevent failures before they occur.

DUO ANALYTICS